

Data Mining for Transportation Mode Recognition from Radio-data

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ABSTRACT

The rapid development of urban information techniques, mobile sensors, and artificial intelligence can help to generate solutions for transportation mode recognition (TMR). The Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge organized at the UbiComp 2021 presents a large and realistic dataset with different activities. Different from the previous three years that aimed at transportation mode recognition from the motion sensors, the goal of this year is to recognize 8 modes of locomotion and transportation (activities) in a user-independent manner based on radio-data, including GPS reception, GPS location, WiFi reception, and GSM cell tower scans. In this paper, our team (We can fly) summarizes our submission to the competition. We first preprocess the data, divide the sample time window and deal with missing values, and then extract 303 features from the given sensor, and finally feed these features into the LightGBM classifier. In the experiment, we utilized the training datasets to train our model and achieved macro-F1 score of 0.665 on the valid datasets.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; • **Hardware** → *Sensor applications and deployments*; • **Theory of computation** → *Design and analysis of algorithms*.

KEYWORDS

Activity recognition; Transportation mode recognition; Machine learning; LightGBM; Smartphone

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1 INTRODUCTION

In recent years, with the ever-increasing computing and perception capabilities of smartphone, the smartphone has almost become an integral part of human life. The application of data collection in smartphone has recently attracted the attention of researchers. Transportation mode recognition is a common but critical task in the human activity recognition study, which provides decision support for urban journey planning [7], public facility arrangement [16], travel route recommendations [14], etc.

The SHL recognition challenge dataset [4, 12] contains multi-modal locomotion and transportation data, which can be used in an activity recognition challenge. Different from the SHL recognition challenge 2018 [13], SHL recognition challenge 2019 [10], and SHL recognition challenge 2020 [11] that aimed at transportation mode recognition from the motion sensors, SHL recognition challenge 2021 focuses on recognizing 8 modes of locomotion and transportation (activities) in a user-independent manner based on radio-data, including GPS reception, GPS location, Wifi reception and GSM cell tower scans. In past competitions, JSI_First [5] derived additional sensor streams from the existing sensors and calculated 858 features in the frequency domain and time domain. They then used cross-location transfer learning via specialized feature selection and performed a two-step classification. Yonsei-MCML [1] proposed a deep multimodal fusion model. The sensor data were independently pre-processed via a convolutional neural network (CNN), and the results were combined with the EmbraceNet fusion algorithm. We-can-fly [15, 17] proposed DenseNetX model working on the multi-channel sensor data simultaneously. Then the concatenated features were fed into the GRU model, and the output features of GRU were used together with the DenseNetX features for the final classification.

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In addition, some researchers who pay attention to transportation mode recognition based on radio frequency signals. In [2], researchers proposed a novel method which also considers the proximity patterns of WiFi and Bluetooth devices in the environment, which were expected to be quite specific to the different transportation modes. Human travel behavior can also be obtained from the trajectory data generated by GPS devices. Wang et al.[8] extracted basic features consist of distance feature, five velocity-related features and two acceleration-related features. Besides, three advanced features including heading change rate, stop rate and velocity change rate with basic features were fed into the LightGBM classifier. Dabiri et al.[3] proposed a deep Semi-Supervised Convolutional Autoencoder (SECA) architecture that can not only automatically extract relevant features from GPS segments but also exploit useful information in unlabeled data.

In this paper, due to the slow sampling frequency of radio-data and the data content has unique semantic information, we divide a large time window to construct the sample, and then extract 303 informative hand-crafted features from GPS, location, WiFi, and cell. Finally, we feed all features into the LightGBM (LGB) [6] classifier. We participate in the SHL recognition challenge 2021 under the team name of “We can fly”.

2 SHL CHALLENGE 2021 DATASET

The SHL recognition challenge dataset contains eight modes of locomotion and transportation, including still, walking, run, bike, car, bus, train, and subway. The 2021 challenge used a subset of the SHL recognition challenge dataset, and all of the participant teams aim to accomplish the task below. The sponsor divided the data into three parts: train, validate, and test. The data comprises of 59 days of training data, 6 days of validation data, and 39 days of test data. The sampling rate is roughly 1 Hz, but is time-varying for each sensor.

The goal of this year is to recognize 8 modes of locomotion and transportation (activities) in a user-independent manner based on radio-data, including GPS reception, GPS location, Wifi reception and GSM cell tower scans. The train data contains the raw sensors data from one user (user 1) with a phone at the Hips location. The validation data and testing data are collected by User2 and User3 with a phone at the Hips location. Both the training and validation splits contain the ground-truth labels along with the sensor data, whereas the testing split does not. The challenge this year is different from the previous three years that aimed at transportation mode recognition from the motion sensors.

3 APPROACH

In this section, we mainly focus on data pre-processing and extracting features from raw data. After that, we feed all features into the LightGBM classifier. First, based on the timestamp of the label, we construct the training and testing sample from the raw sensor data in a sliding window, and then time sequence features and semantic features are manually extracted in various ways from the GPS, Location, WiFi and Cells, respectively. In addition, we design high-order hybrid features for semantic interaction of different sensor information.

Table 1: Statistical features for each sensor

FEATURE	DESCRIPTION
Min	Minimum value of data
Max	Maximum value of data
Mean	Average value of data
Range	Max - Min
Std	Standard deviation value of data
Var	Variance value of data
Median	Median value of data
Quarter	Quantile75 - Quantile25
Autocorrelation	Skewness and Kurtosis

3.1 Data Pre-processing

When the user collects data, all sensor data adopts asynchronous sampling, the sampling rate is about 1Hz. However, the sensor is prone to missing data during the collection process, so it cannot be aligned with the label data. In order to solve the above problems, we fill the missing sensor data by timestamp proximity strategy. Specifically, we use the latest timestamp data to fill in the missing values within a 10-second window, and align the original data samples with the timestamps of the label timestamp. In addition, we observe that the process of collecting data presents a continuous state in terms of timestamps and label switching rarely occurs in a short period of time. To better capture the deeper semantic information of the data, we divide the raw data by the time window size T and compose new data samples. We assume that the original data $D = \{x_1, x_2, \dots, x_n\}$, and n is the sample size of the original data, then the composition of the data divided by the time window size T is as follows:

$$D_W = \{l_1, l_2, \dots, l_i, \dots, l_m\}, \quad m = \lfloor \frac{n}{T} \rfloor \quad (1)$$

$$\text{where } l_i = \{x_{(i-1) \times T + 1}, \dots, x_{i \times T}\} \quad (2)$$

3.2 Features Engineering

Feature engineering plays a very important role in machine learning and often directly affects the effectiveness of the model. We extract 303 features from Location, GPS, WiFi, and Cells data. For a more comprehensive perception of users' transportation mode, we not only extract common statistical features, but also mine deeper semantic features. Table 1 summarizes the set of statistical features computed in this paper.

3.3 Location Feature

The Location dataset has four raw information: Accuracy, Latitude, Longitude and Altitude. Since Altitude is not a good measure of the variation in transportation mode, we only extract features from the other three raw information. There are obvious differences in the moving distance of different traffic modes within a fixed time window. First, to represent the movements trend of users, we calculate the range of the user's movement at the latitude and longitude level within a sample data of time window size T and then convert it into the movement distance in the earth coordinates. In addition, accuracy represents that the accuracy of current location point, and we use the average of all the accuracies collected in a sample to

represent the accuracy of the sample positioning. Furthermore, in order to explore deeper and finer-grained semantic information of the user, we divide the whole time window of T into several samples of time window t again, and then subtract the minimum value on the overall time window T respectively to represent the relative change trend of the user's movement in a short period of time, and calculate a set of statistical features. The specific composition is represented as follows:

$$D_W = \{l_1, l_2, \dots, l_m\} \quad (3)$$

$$D_W = \{r_1, r_2, \dots, r_i, \dots, r_{m-t+1}\} \quad (4)$$

$$\text{where } r_i = \text{Mean}(\{l_i, l_{i+1}, \dots, l_{i+t-1}\}) - \text{Min}(D_W) \quad (5)$$

3.4 GPS Feature

The GPS dataset represents all satellite data collected at each timestamp, including the number of satellites and the original information of each satellite: ID, SNR, Azimuth, Elevation, respectively. First of all, to evaluate the variation of the number of satellites in the data sample over the time window, we calculate the statistical features for the number of satellites that have appeared. Then, in order to characterize the variation of SNR of each satellite, while avoiding the negative effect of repeated appearances of the same satellite within the whole time window, we group all the satellites that appeared within the whole time window by ID and calculated the statistical features. In addition, The satellite azimuth is based on the standard of the direct north direction (about the geomagnetic South Pole). The satellite antenna points to the east or west to adjust an angle, which reflects the position of the satellite in space. Its value ranges from 0 to 360. In order to count the distribution of satellites within the whole time window, we separate the Azimuth and Elevation of satellites by r degrees to make the statistical quantities of binned features. Finally, we also consider the distribution of satellites, to further explore the topological relationship of satellites in spatial, the maximum proportion of the number of satellites within the range of k degrees to the current total number of satellites at each time of the sample separately.

3.5 WiFi Feature

The WiFi dataset contains all WiFi data collected at each timestamp, including the number of WiFi and the raw information of each WiFi: BSSID, SSID, RSSI, Frequency, and Capabilities, respectively. The variation of the WiFi Mac address within the time window can also reflect the user's movement. Firstly, to measure the variation of WiFi in the sample over the whole time window, we calculate statistical features of the number of WiFi that have appeared in the sample. Then, in order to explore the deeper semantic information of WiFi, we calculate the increase or decrease of WiFi number between the first and the last timestamp within the time window. Besides, We perform a weighted average of the above two features to get the overall trend of change. In order to count the RSSI changes of WiFi on the overall time window for the samples, we group the WiFi information by BSSID for the whole time window and calculate the statistical features. Finally, in order to count the RSSI distribution of WiFi over the whole time window, we do binned features statistics quantities with d as the signal interval for them.

3.6 Cells Feature

The Cells dataset indicates all the cell data collected under the current time, and consists of the number of cell and the original information of three categories of cells: LTE, WCDMA, and GSM, respectively. We calculate the similar features for different categories of cells information. In order to statistically represent the variation of cell numbers within the overall time window, we calculate statistical features of the number of cells in different categories separately. Next, in order to represent the distribution of dBm of the cell within the time window, we calculate statistical features for the dBm values of different categories of the Cell within the window, respectively. In the end, to observe the distribution of cell's level within the window, we take different cell's signal level values by doing a binned feature vector.

3.7 Hybrid Feature

In order to mine richer semantic information, we combine the accuracy features of the Location dataset and the number of satellites features of the GPS dataset, and design a function to represent the confidence of this data:

$$y = \log_{\text{number_gps}} \text{accuracy} \quad (6)$$

For missing features or abnormal features, we also designed the following principles to fill in the data:

$$\text{number_gps} = \text{Max}(\text{number_gps}, 2) \quad (7)$$

$$\text{accuracy} = \begin{cases} 25, & \text{number_gps} < 5 \\ 28, & 5 \leq \text{number_gps} < 10 \\ 14, & 10 \leq \text{number_gps} < 15 \\ 8, & 15 \leq \text{number_gps} < 20 \\ 6, & 20 \leq \text{number_gps} \end{cases} \quad (8)$$

At the same time, we transform the distance moved in the earth coordinates calculated in the Location dataset into one-hot vectors according to the following rules, and then multiply them by the confidence level calculated above:

$$\begin{cases} \text{mobile_range} \leq 0 \\ 0 < \text{mobile_range} \leq 20 \\ 20 < \text{mobile_range} \leq 60 \\ 60 < \text{mobile_range} \leq 100 \\ 100 < \text{mobile_range} \leq 150 \\ 150 < \text{mobile_range} \leq 200 \\ 200 < \text{mobile_range} \leq 300 \\ 300 < \text{mobile_range} \leq 400 \\ 400 < \text{mobile_range} \leq 550 \\ 550 < \text{mobile_range} \leq 800 \\ 800 < \text{mobile_range} \leq 1200 \\ 1200 < \text{mobile_range} \leq 1600 \\ 1600 < \text{mobile_range} \end{cases} \quad (9)$$

4 EXPERIMENTS

In our pre-processing data phase, we set the whole time window size T to 90-second. In our features engineering phase, we set the smaller time window size t of Location to 5-second and 10-second

Table 2: Confusion matrix of validation dataset

	1	2	3	4	5	6	7	8
1	22731	1266	10	0	135	1164	1240	3290
2	2286	18267	110	133	338	2781	1792	442
3	0	753	449	900	402	270	0	0
4	108	850	1771	8057	0	0	0	1245
5	600	717	90	810	14213	1193	1620	1230
6	360	174	0	0	0	7535	839	270
7	915	306	0	0	1742	1487	16501	857
8	270	845	0	0	540	150	2398	17506

to extract features respectively, the binning interval degree r of GPS to 15 and the azimuth range k is 90, 180, 270, respectively, and binning interval value of WiFi d is 10. In our model training phase, we set the maximum depth of the tree to 5 and use cross-entropy as the loss function. During the training process, we set a starting learning rate of $5e-3$. In our parameter tuning phase, we leverage train dataset to train our model and valid dataset to validate our model. Each training includes a total of 15000 epochs, and we select the model parameters corresponding to the lowest loss in the valid datasets from 15000 epochs as the final model. During our commit phase, we trained with all the data to improve the performance of the model.

4.1 Experimental Results on Valid Datasets

In the previous SHL recognition challenge, F1-score is adopted for evaluation metric. Therefore, we evaluate our model using the F1-score method by the valid datasets. We obtained F1 scores of 0.665 on the valid datasets. Besides, Tables 2 show the confusion matrix on the valid datasets. The 8 class activities are: 1 - Still; 2 - Walk; 3 - Run; 4 - Bike; 5 - Car; 6 - Bus; 7 - Train; 8 - Subway. From table 2, we obtain the following conclusions. The class Run is the most difficult activity to recognize, followed by Walking, Bike and Car. It can be observed from the confusion matrix that, Run tends to be misclassified as Walking and Bike. The motion of the smartphones during Train, Bus and Subway is more distinctive than the transportation modes that rely on user's movement.

4.2 Computational Resources

We used 1 CPU on 1 server.

- 1 CPU Intel(R) Xeon(R) Gold 6132 CPU, 128GB RAM
- Language: Python 3.8.0
- Framework: LightGBM 3.2.1

The size of the trained model is about 80 MB. It took to train the model for about thirty minutes. It took to evaluate the test dataset for about 1 minutes.

5 CONCLUSION

We proposed a classical pipeline for the SHL recognition challenge 2021. We construct 303 features from GPS, location, WiFi, and cell. We not only consider statistical features, but also semantic feature, as well as feature crosses. Then we feed the features into LGB. After soft-max, we got the final classification result. As a

result of classifying test data of the SHL recognition challenge, our model obtained macro-F1 scores of 0.665 on the valid datasets. The above results indicate that the missing value has a great influence on transportation mode recognition. The recognition result for the testing dataset will be presented in the summary paper of the challenge. [9]

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